

Optimizing Intraday Breakout Strategies on the NSE: A Block-Based Performance Evaluation

WANG CHENXI - wcx0209@hotmail.com

SIDDHANT GANGWAR – sidd.0ct7@gmail.com

This file includes a link to online python notebooks

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Abstract

Intraday trading is playing a key role in today's financial market in mitigating overnight risk and capturing short-run movements in prices in each trading day. In emerging economies such as India's National Stock Exchange (NSE), two key intraday phenomena—high volatility at market opening and rising volumes towards market close—make for conditions favorable for Opening Range Breakout (ORB) strategies. The paper in hand considers alternative ORB setups (time frames of 5, 15, and 30 minutes, threshold surge volumes at 1.2× and 1.5×, and holding periods of N=2, 3, and 5) based on Tata Motors intraday prices. ORB daily returns are compared to a simple Buy & Hold (BH) strategy, in which BH is buying at daily open and selling at daily close.

Our method is a direct ORB–BH daily difference, compared using a bootstrap method to obtain p-values. ORB variants across the board have larger cumulativereturns than BH (often by significant margins), but p-values tend to be in the range 0.45–0.50, suggesting there is insufficient statistical evidence to claim ORB's advantage is better than random luck. A block-by-block breakdown pseudo–out-of-sample split better illustrates ORB raw outperformance in multiple chunks but none produce lower p-values. Bull vs. Bearish breakout directions are compared as well. In each, outcomes have strong practical returns but are still statistically indecisive in a one-year sample for one security. We conclude ORB is operationally appealing but would need stricter data coverage, transactions cost modeling, and advanced significance testing to confirm or target in on apparent advantage in this strategy.

1. Introduction

Intraday closing out daily positions or, in general parlance, intraday trading, reduces risks related to overnight news or announcements at the macro level. As opposed to holding instruments overnight for uncontrollable after-market events, intraday players close out at session close, thereby reducing exposure to only intraday volatility. The National Stock Exchange (NSE) in India is a very interesting context for intraday strategies in view of two prominent features:

1. High Market Open Volatility: The interplay of overnight news releases, pre-market orders, and global market sentiment can combine and produce a rapid period of price discovery in opening market minutes.
2. High Volume Near Market Close: Different types of institutional rebalancing and fund transactions concentrate at or near close, increase order flow and yield reversal or late-session volatility.

Such intraday characteristics are favorable for breakout strategies in which players are anticipating capturing a rally in price after it breaks past a previous-established level. The Opening Range Breakout (ORB) system, as formulated and presented by technical analysts (Edwards & Magee, 1948; Crabel, 1990), is specifically targeting early in the day price range—the start 5, 15, or 30 minutes in trading time—and is waiting for the price to break past established high or low and thereby trigger a potential bull or bear move, respectively.

1.1 Motivation and Research Problem

Though ORB strategies are easy in theory—"short when it breaks below, buy when it breaks above opening range"—their actual performance can be highly market- and volatility regime-dependent and subject to additional filters such as confirmation based on volumes. ORB-based strategies are extensively used in retail trading but there is relatively little peer-reviewed research in Indian markets examining statistical significance compared to naive Buy & Hold (BH) in short intraday time frames.

We try to close this research gap by testing rigorously:

- **Multiple ORB time windows** (5, 15, 30 minutes).
- **Volume spike filters** at 1.2× and 1.5× daily average volume.
- **Different holding periods** (N=2, N=3, N=5 bars) to see how quickly trades are closed out.
- **Block-by-block** pseudo-out-of-sample slicing to test consistency across different sub-periods within our dataset.
- **Separate bullish vs. bearish** signals, as negative momentum in a downtrend can behave in a different way than positive momentum in a rising trend.

1.2 Overview of Methods and Findings

We measure performance as a day-to-day difference in ORB relative to a daily benchmark for BH (buying at market open and selling at market close). By assigning ORB no-trade days a difference in returns of zero, we have an apples-to-apples daily basis for comparison to BH. Then we do a bootstrap on the daily difference distribution in order to have p-values for testing whether difference in means could readily occur as random fluctuation.

Even though total raw outperformance at the end is much better in ORB (ending at +20% to +30% vs. a ~-36% fall for BH), p-values remain fairly high (often close to 0.50) and hence we can't affirmatively claim advantage at typical confidence (say, 5% alpha) level based on strict statistics. The difference is in line with practical observation (ORB line much above BH in final values) but in

opposition to statistical significance (probability for random daily volatility to replicate like discrepancies on repeated sampling).

2. Literature Review

2.1 Market Microstructure and Intraday Behavior

Market structure research formulates theoretical frameworks for how prices move in a day (Kyle, 1985). Intraday opening volatility is a result of overnight information concentration, and at close there are significant institutional trades pushing volume spikes (Hasbrouck & Saar, 2013). The daily cyclical pattern creates time frames in which enlargements in prices, or breakouts, are more likely to be pivotal.

2.2 Breakout Strategies

Breakout trading is rooted in the principle that when price penetrates a level (resistance, opening range, or support), extended directional movement is in store. Bollinger (2002) introduced volatility-based break-out bands and break-outs across lines of technical support and resistance have been discussed in Murphy (1999). The ORB system (Crabel, 1990) is specifically examining opening minutes in the day for high and low, theorizing moving beyond them as a signal for a strong directional thrust for the remaining portion of the session. Brock, Lakonishok, & LeBaron (1992) demonstrated, however, that such performance based on technical rules can be transitory or market-dependent.

2.3 Volume Confirmation

Volume is utilized as a filter in discriminating between strong breakouts and "noise." According to Granville (1963), volume consolidates movement in prices as being valid, while Pring (2002) depicted how sudden increases in volume are typically linked to strong movements in momentum. Alternatively, if a threshold is too strict, there is a possibility of missing out on good trades (Harris, 2003). On the NSE, Banerjee & Banerjee (Year) depicted how there is increased volume towards closing time, at times leading to further breakouts.

2.4 Intraday Risk Management and Evaluation

Neely et al. (1997) and Krauss et al. (2017) cover measuring raw and risk-adjusted returns (Sharpe, Sortino, drawdown) as being central. Day-by-day difference and related p-values based upon bootstrapping are our focus, though. The direct difference method is central in ascertaining if outperformance is regular enough to be in excess of random luck.

2.5 Block-by-Block Slicing

Block-based performance testing (Brock et al., 1992; Pardo, 2008) partitions information into time-based units. If there is some measure of robustness if the strategy is better in each block, p-values can be large if there is lots of variability day to day since each block is based upon fewer information points.

2.6 The Gap in NSE Studies

Though there are lots of intraday strategies in proprietary trader and forum communities, there is scant serious research-based literature for ORB strategies based on NSE, specifically short time horizons and volume-filtering strategies. The paper makes an attempt to fill this gap and adopt a systematic methodology in correlating practical returns and a statistical viewpoint.

3. Methodology

3.1 Data Acquisition and Preprocessing

We use **Tata Motors** (an NSE-listed stock) at 5-minute intervals for approximately one year. Each row has [time, open, high, low, close, volume]. We:

1. Convert time to a Python datetime object.
2. Derive Date from time (e.g., time.dt.date).
3. Group data by Date to isolate daily sessions, sorting chronologically by time.
4. Remove duplicates or missing intervals if encountered.

This single-instrument, single-year limitation frames the scope of our study: it provides a controlled dataset but may not capture all market cycles or cross-sectional diversity.

3.2 Opening Range Breakout (ORB)

For each traded day, we have a given ORB window (i.e., 5 minutes). That is:

- ORB_High = peak high in opening X mins in a day.
- ORB_Low = lowest in the first X minutes of the day.

Breakout occurs when the (we consider close in each bar) price is above ORB_High (upside breakout) or below ORB_Low (downside breakout). On breakout, we look at returns for next N bars ($N=2,3,5$). If there is a second breakout in a day (i.e., if there is some time when price is bullish and later is bearish), we ignore it in general in order to avoid excessive transactions and look at just breakout for each day. If no breakout is noticed, ORB daily return is taken as 0%.

3.3 Volume Filters

We optionally require breakout bar volume $\geq$ threshold $\times$ average_volume_for_that_day, for threshold $\in \{1.2, 1.5\}$. As a simple example, if day's average_volume is 100,000, a 1.2 $\times$ filter would need breakout bar volume $\geq 120,000$. It is intended to avoid fake-out momentum, but risks avoiding potentially good trades if close to average volume.

3.4 Buy & Hold (BH) Benchmark

For each day, it buys at the day opening and sells at the day closing, producing one daily return:

$$BH_daily = \frac{close_{end}}{open_{start}} - 1.$$

We repeat this for each day, compounding for the entire sample. Next, we compare ORB daily returns to BH on a daily basis.

3.5 Assigning 0% Return on No-Trade Days

In instances when ORB does not break out on a given date, we have ORB_daily as 0% for the day. Alternatively, BH is +X% or -Y%. It provides us with a difference for the day as (0% - BH%). On a given sample period, each day thus has a given ORB return. It makes us able to measure fairly day-by-day differences.

3.6 Day-by-Day Difference and Bootstrap

For each date i :

- ORB_return_i = intraday breakout return (or 0%).
- BH_return_i = daily buy-hold return.
- $Diff_i = ORB_return_i - BH_return_i$

We collect all $\{Diff_1, Diff_2, \dots\}$. The mean of them, D_{obs} , is a measure of outperformance daily on average. A bootstrap method re-samples them (with replacement) and asks how often a random pseudo-sample mean is at or above D_{obs} . If it is fairly typical, p-value is close to 0.5, and we have no strong significance. If it is unusual (<5% or thereabouts), we have reason to say the difference in the strategy is significant.

3.7 Block-by-Block Analysis

We split the entire year's information in 4 time frames (each time block ~3 months). We perform the above ORB identification in each time block, compute daily variations and block-level p-values. The pseudo-out-of-sample method is applied to ensure if the strategy is reproducible in short sub-periods.

3.8 Bullish vs. Bearish Breakouts

We split daily returns for bull signals and for bear signals out separately. On break days we consider each daily return as the entire day's "bullish trade." The same for bear. Then we compute each subset's daily difference from BH and see if one is more stable or better-performing.

3.9 Plotting Cumulative Returns

We produce **one chart per section**:

- **Section 4.1:** Price-only ORB (5, 15, 30 min) plus BH, all on one figure.
- **Section 4.2:** Volume filters (1.2x, 1.5x) plus BH.
- **Section 4.3:** Holding periods (N=2, N=3, N=5) plus BH.
- **Section 4.4:** Each block's ORB line vs. BH line.

- **Section 4.5:** Bullish ORB line, Bearish ORB line, and BH line.

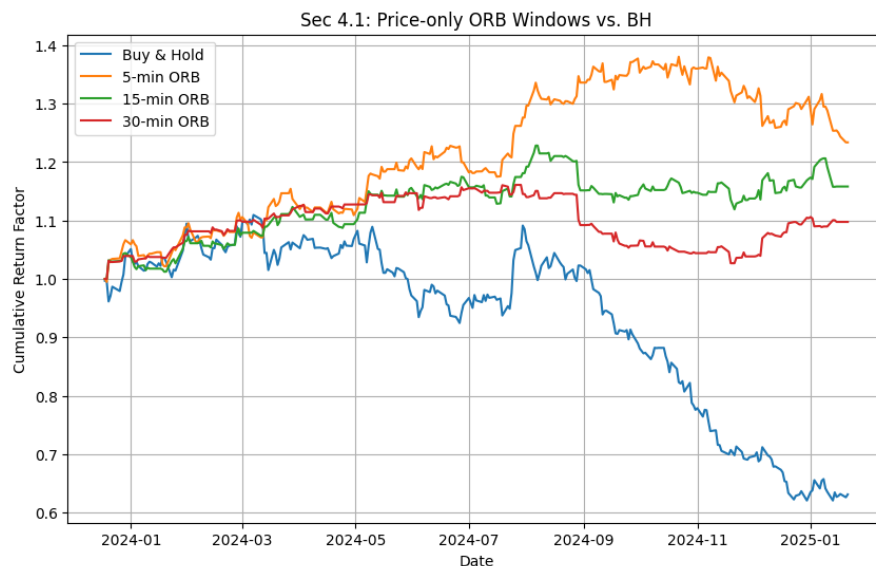
They capture how each variant's total return factor evolves over time. A line for ORB way out in front of BH would normally represent raw outperformance in magnitude but is checked for significance using difference-based p-value.

4. Empirical Results

In this section, we explore how each variation in strategies behaves in real-time. Each figure (plotted as "Sec 5.X" graphs) presents us with a plot of cum. ret. for the sample period. The concluding tables, in turn, report each situation daily difference mean, p-value, ORB vs. BH final cum. ret., excess gain, and days. The database is approximately one year (early 2024-early 2025) and is focused on Tata Motors.

Section 4.1: Price-Only ORB (5-, 15-, 30-Min)

1. Plot:



- As shown in "Sec 4.1: Price-only ORB Windows vs. BH," the BH line declines severely from ~1.0 factor near the start to ~0.60–0.65 by the end, reflecting about –36% net loss.
- The 5-minute ORB line climbs above 1.3 factor by mid-late sample, the 15-minute near 1.2–1.25, and the 30-minute near 1.1–1.15.

2. Statistics:

- **Daily Mean Diff** ~0.19–0.24%.
- **p-value** ~0.49–0.50. This indicates the difference in daily returns might easily emerge from random re-sampling.

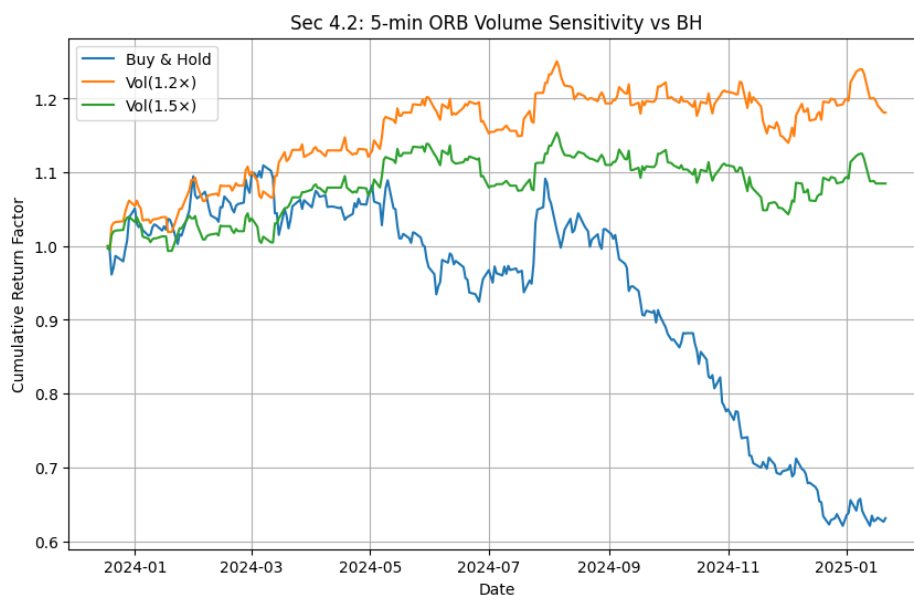
- **Cumulative Return:** e.g., +23.37% for 5-minute ORB, +15.81% for 15-minute, +9.76% for 30-minute, contrasted with −36.88% for BH.

Interpretation:

- From a practical viewpoint, 5-minute ORB is far larger than BH's closing figure. The gap in closing return (approximately 60 percentage points difference) is remarkable.
- On a statistical basis, $p \sim 0.5$ indicates no strong proof that this advantage is "significantly different from zero" in a strict sense. The daily returns are sufficiently unpredictable or the sample is too short, and thus such outperformance is not highly unusual in random re-samplings.

Section 4.2: Volume Spike Sensitivity (1.2× vs. 1.5×)

1. Plot:



- “Sec 4.2: 5-min ORB Volume Sensitivity vs BH” presents lines marked as “Vol(1.2×)” and “Vol(
- The 1.2× is still well above 1.2 multiple for much of the year, and 1.5× is lower but still far above BH's steep dip below 0.70.

2. Stats:

- Daily Mean Diff $\sim 0.19\text{--}0.22\%$.
- p-value $\sim 0.48\text{--}0.50$.
- Cumulative: The 1.2× method yields $\sim +18\%$ versus BH at -36.88% , and the 1.5× method yields $\sim +8\%$

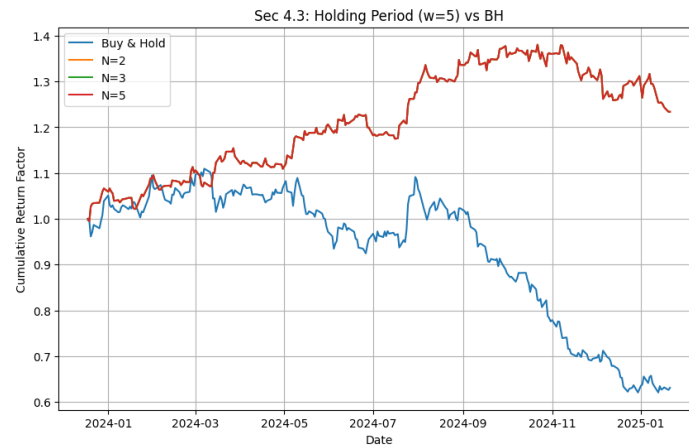
Interpretation:

- Volume reducers reduce total transactions but keep the final outcomes of the plan in good form.

- Again, p-values near 0.50 indicate that at a statistical level, the difference is not "rare." The raw difference is vast, but variability in the sample would appear to allow random subsets to achieve as vast a difference.

Section 4.3: Holding Period Variation (N=2, N=3, N=5)

1. Plot:



- “Sec 4.3: Holding Period (w=5) vs BH” depicts lines for N=2, N=3, N=5, plus BH. All ORB lines converge near 1.3–1.35 factor, while BH ends near 0.60–0.65.

2. Stats:

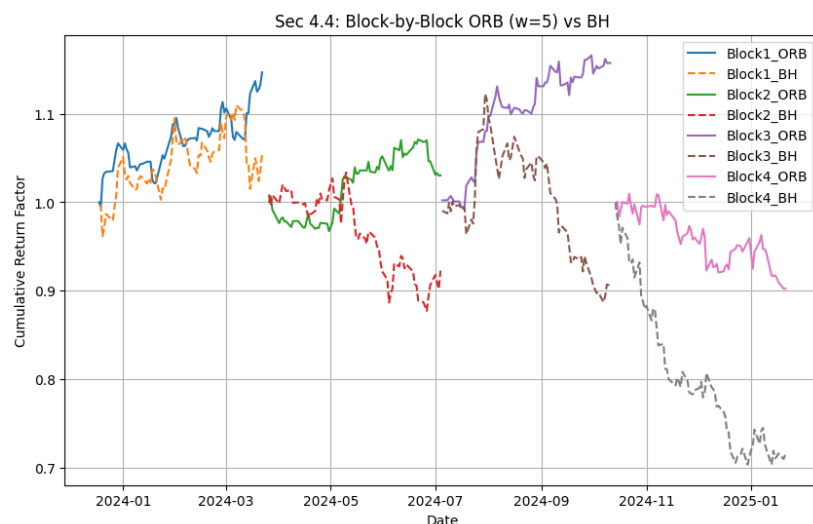
- **Daily Mean Diff** ~0.24%.
- **p-value** ~0.46–0.50.
- **Cumulative:** ~+23–24% for ORB vs. –36.88% for BH, net ~60% difference.

Interpretation:

- Intriguingly, the slight changes in holding bars (N=2 vs. N=5) do not dramatically alter final performance in this dataset—likely because most breakouts complete their moves quickly or fail, so a few extra bars rarely changes the outcome.
- p-values remain ~0.5, reinforcing the theme that the *observed advantage* might not be statistically robust under daily difference re-sampling.

Section 4.4: Block-by-Block

1. Plot:



- “Sec 4.4: Block-by-Block ORB (w=5) vs BH” splits the

year into four roughly equal segments. Each block's ORB line is plotted separately, along with a BH line. Because the blocks are disjoint in time, each line only covers its designated date range.

- The first block sees ORB rise from ~ 1.0 to ~ 1.15 or so, BH from ~ 1.0 to ~ 1.05 . The second block sees ORB from ~ 1.0 to ~ 1.03 , BH from ~ 1.0 to ~ 0.92 , and so on.

2. Stats:

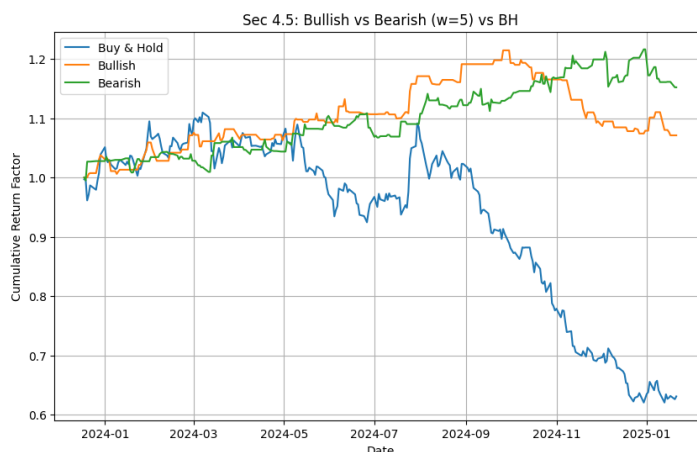
- In each block, **ORB** surpasses **BH** by 8–25% in final cumulative terms.
- p-values range ~ 0.45 – 0.49 , again indicating no block demonstrates a difference that the bootstrap test deems “unlikely.”

Interpretation:

- Block segmentation clarifies that ORB outperforms BH in every sub-period, even when the market as a whole declines significantly. Nonetheless, each block's daily returns remain volatile, producing a high p-value.

Section 4.5: Bullish vs. Bearish

1. Plot:



- “Sec 4.5: Bullish vs Bearish ($w=5$) vs BH” has three lines: BH, Bullish, Bearish.
- The Bearish line tends to be longer than the Bullish line, indicative of short breakouts down being more profitable in this population.

2. Stats:

- Bullish close $\sim +7\%$ vs. -36.88% BH, Bearish close $\sim +15\%$
- p-values ~ 0.48 – 0.49 .
- Initiates approximately 141 short trades and 101 buy trades, and hit ratios at nearly 50–55%.

Interpretation:

- The implication is thus that in a market with a strong negative year-long drift (like in BH's downtrend), short break-out capture is better than bull subset. Again, daily difference dispersion is sufficiently wide for significance not to be established.
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5. Discussion

5.1 Practical vs. Statistical Contradiction

In each period, raw total returns for ORB outstrip BH by enormous spreads (40–60% difference in total returns is not unusual). Still, p-values in the vicinity of 0.45–0.50 persist, suggesting random pseudo-samples replicate the difference in means sufficiently often. One possible explanation:

1. Daily Volatile Returns: The difference series can experience extreme negative or positive volatility in multiple instances.
2. Small Sample: A year or a year and a quarter for one company may not yield sufficient stable signals to eliminate random fluctuation.
3. Correlated or Overlapping Returns: Traditional bootstrap resampling is not able to capture adequately day-to-day autocorrelation or market structure stages.

5.2 Volume Threshold and Holding Period Insights

- Volume: A threshold at 1.2× is observed to yield better absolute performance than at 1.5× since fewer trades are being excluded. However, they are much better than BH's negative closing value.
- Holding period: The consistency in N=2, N=3, and N=5 reflects breakouts usually finish their movements in a short time and thus total bar counts don't significantly affect final outcomes.

5.3 Block-by-Block Consistency

We dissect the data and ORB is still outperforming in each block-guaranteeing some raw outperformance consistency. Each block in individual form does not, though, yield a highly significant difference as p-values are still relatively high. The method is indicative in nature in that good raw performance does not necessarily translate to a p-value being small when daily volatility and associated trends are still relatively high.

5.4 Bullish vs. Bearish

Bear indicators modestly outpace Bullish in closing returns, capturing that Tata Motors or market trend in general is negative for much of our sample. Still, no directional subset yields a p-value lower than 0.45, and we can't conclude shorting breakouts is inherently profitable.

5.5 Potential Explanations for High p-values

1. Non-Normal Return Distribution: Daily differences may have heavy tails and thus simple bootstrap may fail to capture the tails of the distribution.

2. Short on Cyclical or Regime Diversity: A year can be short on cyclical or regime variety.
 3. Daily Correlations: Consecutive days can have carryover or exogenous macro influence, creating increased sampling variability.
 4. Zero Cost Factor: Actual trading slippage/fees may lower net profits, potentially closing the ORB–BH gap still further.
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6. Limitations and Future Research

1. Data Scope: A single stock for one year, though neat in presentation, greatly limits broader generalizations. Extension to multiple NSE stocks or multiple years could yield lower p-values if there really is a trend.
 2. Transaction Cost and Slippage: ORB can be traded multiple times daily or require multiple partial closing for the day. Effective net returns can be depleted once we account for brokerage charges, transactions charges for exchanges, and potential slippage (the difference between theoretical fill and actual fill prices).
 3. Correlation-Aware Testing: The daily returns are assumed to be serially correlated (specifically for trends in multiple days in a market). More advanced randomization or sign-flip testing allowing for autocorrelation may yield different significance results.
 4. Machine learning or Dynamic Filtering: Instead of hard-coded 1.2×, 1.5× volume thresholds, there may be volatility-based thresholds or transitory trend-based thresholds.
 5. Strong Out-of-Sample: Since block-based split is only an out-of-sample estimation, a separate holdout database or rolling forward-walk analysis would be able to validate if these results generalize outside of the immediate time period.
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7. Conclusion

7.1 Summary of Findings

Our thorough review of Opening Range Breakout strategies applied to Tata Motors intraday prices gives us:

- Time Window (5, 15, 30 min): Each gives gigantic final profits beyond BH, and the 5-minute method is typically largest.
- Volume Threshold (1.2×, 1.5×): They yield good outperformance, but 1.2× is better in eventual performance.
- Holding Period (N=2, 3, 5): Small difference in final performance since breakouts are going to succeed or fail in a short time period in bars.
- Block-by-Block: Even raw performance across each block, without anybody demonstrating a difference at typical confidence thresholds.

- Bearish vs. Bullish: Bearish strategies are slightly better than bullish strategies, reflecting the general downtrend in BH's eventual performance.

7.2 Practical vs. Statistical Duality

Although raw outcome ORBs continue to dominate (ORB lines are holding up much better than BH in each plot), daily difference method yields p-values at about 0.45–0.50—i.e., no good evidence daily superiority is beyond random variation. It shows a main principle in quantitative finance: practical significance in real returns in a finite population is not necessarily "statistical proof" for alpha, if daily variation is still prominent or if the population is finite.

7.3 Implications for Practitioners

In actual market trading, an ORB-based system would be viable if it is resistant to transaction costs, or if real-time surges in volumes confirm strong trends. The repeat advantage across different sets of parameters signifies feasible operability if real fill prices and slippages are managed. Caveats are still necessary: longer history in the data, real-time slippage estimation, and significance testing for correlations are recommended before implementing at scale.

7.4 Future Work

- Expand to More NSE Stocks: Cross-testing in multiple liquid stocks can validate if ORB outperformance is general or is limited to Tata Motors' idiosyncratic volatility behavior.
- Transaction Cost Modeling: The inclusion of brokerage fees, taxes, and potential slippage may uncover lower net profits.
- Sophisticated Machine Learning Filters: Adaptive networks or gradient boosts capable of adapting volume thresholds or sensing intraday microstructure dynamics would be able to enhance breakout signals.
- Strong Significance Testing: From advanced bootstrap or sign-flip testing for dealing with time-series autocorrelation, to bull/bear cycle-capturing longer-term expansions, later research can target ORB performance based on better statistical testing.

In general, ORB-based strategies on the NSE, as exemplified for Tata Motors, have vast raw outperformance relative to naive open-close holding. Yet, nearly half-likelihood p-values affirm the possibility for this advantage not being established at conventional significance levels in the constraints of a one-year dataset and difference methodology on a day-by-day basis.

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Author's Profile

Name: WANG CHENXI

Affiliation: MFE 2024, WORLDQUANT UNIVERSITY

Email: wcx0209@hotmail.com

Name: SIDDHANT GANGWAR

Affiliation: MFE 2024, WORLDQUANT UNIVERSITY

Email: sidd.0ct7@gmail.com

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